NTI Academic Success & Intervention

Student Academic Health Analysis — Project Report

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Audience	ENG. DONIA — Data Analysis Evaluation Committee
Period	March 2024 – March 2025
Stack	SQL Server 2025 → Python 3.12 → Power BI

1. Executive Summary

This report documents a full end-to-end data analysis project examining the academic health of 12,156 students across five grade levels over a full academic year. The project follows a structured three-phase pipeline: SQL storytelling to explore and document the raw data, Python cleaning and feature engineering to prepare it for analysis, and a six-page Power BI dashboard to surface actionable insights for school leadership.

The central question driving every phase of this project is simple: which students are at risk of academic failure, and what combination of behavioral signals identifies them earliest. The answer required integrating five separate data sources — attendance, homework, performance, student profiles, and parent communication — into a unified analytical view.

Key Metrics at a Glance

Metric	Value
Total students	12,156 across Grade 1–5
School average exam score	74.96
At-risk students identified	~2,000 (12.41%)
Overall attendance rate	25.12% (present only)
Homework completion rate	49.99%
Silent students (no communication)	1,644 (13.5%)
Exam score range	40 – 110
Data period	March 2024 – March 2025

2. Project Overview

2.1 Objective

Design and deliver a complete data analysis pipeline for a school dataset — from raw CSV files to an interactive Power BI dashboard — that enables school administrators to identify at-risk students, monitor attendance and engagement trends, and evaluate the effectiveness of parent communication.

2.2 Pipeline Architecture

The project is organized into three sequential phases, each building on the output of the previous one.

Phase	Tool	File	Output
1	SQL Server 2025	storytelling.sql	Documented insights + quality flags
2	Python 3.12	education_cleaning.ipynb	5 clean CSVs + student_summary.csv
3	Power BI	Final Education.pbix	6-page interactive dashboard

2.3 Dataset

Table	Rows	Description
students.csv	12,156	Student profiles — Grade 1 to 5
attendance.csv	364,680	Daily attendance per subject
homework.csv	60,780	Assignment status and guardian signatures
performance.csv	36,468	Exam scores and homework completion rates
teacher_parent_communication.csv	24,312	Parent-teacher message logs

3. Phase 1 — SQL Storytelling

The SQL phase is structured as a narrative of six analytical chapters. Each chapter answers a specific business question rather than performing generic data exploration. All findings and data quality issues discovered during this phase were documented as comments directly inside the script for full traceability.

3.1 Chapter Structure

#	Chapter	Analytical Focus
1	Who are the students?	Population and grade-level distribution
2	Are they showing up?	Attendance patterns, monthly trends, subject breakdown
3	Are they doing their work?	Homework compliance, guardian signature correlation, grade feedback
4	Are they performing?	Exam score distribution, subject and grade-level averages
5	Are parents in the loop?	Communication volume, message type split, silent students
6	At-risk student profile	Cross-table query: attendance < 20%, score < 60, missed HW > 3

3.2 Data Quality Issues Identified

All issues discovered were flagged in SQL comments and resolved in the Python phase. UPPER() and TRIM() were applied inline during SQL analysis only to ensure query accuracy — no permanent changes were made at this stage.

Column	Issue
students.Date_of_Birth	Two coexisting formats: YYYY-MM-DD (10,951 rows) and MM-DD-YYYY (1,205 rows)
attendance.Attendance_Status	Inconsistent casing across values, leading whitespace, and typo 'absnt' for 'Absent'
homework.Status	Six variant representations mixing emojis (checkmark, X) with text labels
homework.Guardian_Signature	Imported as BIT type by SSMS; blank strings treated as a third category instead of NULL
performance.Homework_Completion_%	7,376 records contain -5 (logically invalid) and inconsistent % suffix across values

4. Phase 2 — Python Cleaning & Feature Engineering

The Jupyter Notebook processes all five tables in sequence, resolves every data quality issue flagged in the SQL phase, validates referential integrity across all tables, engineers the features needed for the Power BI dashboard, removes unused columns, and exports five clean CSV files.

4.1 Cleaning Decisions

Issue	Decision	Justification
Mixed date formats	pd.to_datetime(format='mixed')	Handles both formats without ambiguity
Attendance casing + typos	str.strip().str.upper() + map	ABSNT merged into ABSENT
Homework status variants	Set-based function mapping	Done | Not Done | Pending — clean 3-label taxonomy
Guardian_Signature nulls	fillna(False)	No evidence of sign-off = treat as unsigned (conservative)
Homework_Completion_% nulls	Per-subject median imputation	Subject medians differ (Arabic 95, English 90) — global mean introduces bias
Homework_Completion_% = -5	Replace with NaN	Logically invalid — no completion rate can be negative

4.2 Feature Engineering

A student-level summary table (student_summary.csv) was built as the primary fact table for the Power BI dashboard. Each row represents one student and aggregates all behavioral signals.

Feature	Source	Description
Attendance_Rate_%	attendance	% of sessions where status = Present
HW_Completion_%	homework	% of assignments with status = Done
Avg_Exam_Score	performance	Mean exam score across all subjects
Missed_Assignments	homework	Count of Not Done assignments
Comm_Count	communication	Total parent-teacher messages for this student

Has_Communication	derived	Boolean — True if Comm_Count > 0
Is_At_Risk	derived	True if Attendance < 20% AND Score < 60 AND Missed > 3
Score_Band	derived	Failing / Average / Good / Excellent based on Avg_Exam_Score
Grade_Num	students	Ordinal integer (1–5) for Power BI grade sorting

4.3 Columns Removed

Columns not required by any dashboard page were dropped before export to reduce file size and avoid exposing unnecessary data.

Table	Column	Reason
students	Emergency_Contact	Not used in any dashboard page
students	Date_of_Birth	Replaced by derived Age column
attendance	Year	Month already contains year and month information
homework	Assignment_Name	Too granular for any dashboard visual
performance	Teacher_Comments	Free text — not usable in visual analytics
communication	Message_Content	Free text — not usable in visual analytics

5. Phase 3 — Power BI Dashboard

The dashboard consists of six thematic pages, each answering a distinct analytical question. All pages share a consistent design language with Quick Navigation buttons for cross-page movement and a Grade Level slicer for filtering.

5.1 Page 1 — Executive Summary

Provides a school-level health overview. KPI cards show Total Enrollment, Average Academic Score, High-Risk Cohort count, Attendance Rate, and Homework Completion Rate. A donut chart breaks down the score band distribution and a bar chart shows at-risk count by grade. A monthly attendance trend line closes the page.

5.2 Page 2 — Attendance Analysis

Breaks down all five attendance statuses (Present, Absent, Late, Left Early, Excused) at the subject and grade level. Charts include a stacked bar showing status distribution by subject, a

bar showing average attendance by grade, and an absence and tardiness breakdown by subject. Date range slider enables time-based filtering.

5.3 Page 3 — Homework & Engagement

Focuses on student effort. KPIs show Pending Rate, Missed Assignments Rate, and Completion Rate. Two donuts show overall completion status and completion rate per subject. A horizontal bar ranks grade feedback distribution from A+ to C-.

5.4 Page 4 — Academic Performance

Examines exam results. KPIs show score range (40–110), High Achievers percentage, Failing Rate, and Average Score. A bar chart ranks students by performance band. Average score is shown by both subject and grade level. A histogram displays the full score distribution in bins.

5.5 Page 5 — Parent Communication

Analyzes the school's communication coverage and behavior. KPIs show total Communication Volume and Silent Students count. A donut breaks down message types (Teacher to Parent, Parent to Teacher, Automated Reminder). A bar shows silent student distribution by grade. A multi-line trend tracks daily communication volume across all three message types.

5.6 Page 6 — At-Risk Intervention

The action page. KPIs show the at-risk cohort's average score, average attendance, at-risk percentage, and total count. A scatter plot maps every student by Attendance Rate versus Exam Score, colored by at-risk status. A full student-level table lists each at-risk student with all relevant metrics for direct intervention prioritization.

6. Key Findings

6.1 Student Population

- 12,156 students are distributed near-equally across five grade levels, each holding approximately 20% of the total population.
- The balanced distribution means performance differences across grades reflect genuine behavioral or instructional variance, not sampling imbalance.

6.2 Attendance

- Only 25.12% of all attendance records are marked Present — students are physically present in roughly 1 in 4 tracked sessions.
- Late arrivals account for an additional 24.96% of records, making tardiness as common as full absence.
- Absent and ABSNT (a typo variant) together account for ~25% of records. These were unified during the Python cleaning phase.
- No significant difference in attendance rates exists across grade levels or subjects — absence is general, not subject-specific.

6.3 Homework

- Overall homework completion rate is 49.99%. Approximately 1 in 3 assignments is not completed.
- Science has the lowest completion rate (49.51%). Geography is highest (50.45%). The spread is narrow across subjects.
- Grade feedback is uniformly distributed across all eight grade bands (A+ through C-), each at approximately 12.5%.

6.4 Academic Performance

- The school-wide average exam score is 74.96 across a valid range of 40 to 110.
- 12.87% of students are classified as High Achievers. The failing rate stands at 12.41%.
- Average scores are nearly identical across all subjects (range: 69.85 to 70.25) and across all grade levels (range: 69.81 to 70.24).

6.5 Parent Communication

- 1,644 students (13.5% of the school) have zero parent communication records across the full year.
- Grade 1 has the highest silent student rate at 14.73% — the most critical finding in this chapter given the age and dependency of these students.
- Communication peaks in December and January, consistent with exam season behavior. March 2025 data is a partial month.
- Students with active communication score only marginally different from silent students (69.90 vs 69.97), suggesting communication volume alone is not sufficient — quality and follow-through matter more.

6.6 At-Risk Students

- Approximately 2,000 students (12.41% of the school) are flagged as at-risk based on three simultaneous conditions: attendance below 20%, average exam score below 60, and more than 3 missed assignments.
- The majority of at-risk students have parent communication records. This means contact is happening but not translating into improvement — the intervention strategy, not its frequency, needs to change.
- Students with no parent contact and failing indicators represent the highest-risk tier requiring immediate action.

7. Recommendations

7.1 Immediate Actions

- Deploy the At-Risk Intervention page as a working tool for grade supervisors. Each row in the table is a student requiring a direct conversation before the academic year ends.
- Establish a Grade 1 parent onboarding protocol. A 14.73% communication gap at the entry grade is a systemic issue that needs a structural fix, not ad-hoc outreach.
- Review the communication strategy for the ~200 at-risk students who have active contact but declining performance. Volume is not the problem for this group.

7.2 Data Quality Improvements

- Standardize the Attendance_Status field at point of entry. The ABSNT typo and casing inconsistencies indicate manual data entry without validation.
- Replace emoji-based homework status encoding with a dropdown selection in the source system. Emoji values are fragile across database imports and require manual mapping on every pipeline run.
- Set a hard constraint on the Date_of_Birth field to enforce a single date format. The mixed ISO/US format causes silent parsing errors that make age analysis unreliable.

7.3 Next Pipeline Iteration

- Add a time-series attendance model that flags declining attendance trends for individual students before they cross the at-risk threshold.
- Build a subject-level performance model that identifies whether a student's low score in one subject is isolated or part of a broader academic decline.

8. Repository Structure

File / Folder	Description
data/raw/	Original five CSV files as received
data/clean/	Five cleaned CSVs exported from Python phase
storytelling.sql	Phase 1 — six-chapter SQL storytelling script
education_cleaning.ipynb	Phase 2 — Python cleaning and feature engineering notebook
Final Education.pbix	Phase 3 — six-page Power BI dashboard
README.md	Visual-first project overview for the GitHub repository